Intellectual Property & Privacy (IPP)

JM0160



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Lecture 6: Database Protection

Recap of Knowledge Clips 6.1. & 6.2

You should understand:

- KC 6.1: What is the subject-matter, criteria for protection and scope of protection of EU copyright law.

- KC 6.2:
  - How is a database defined under the Database Directive (1996).
  - What is the difference between copyright and the sui generis database right?

Any questions?

Subject-matter

Article 1(2) DD: Definition of a database (covers both electronic and non-electronic databases).

A **collection** of **independent** data arranged in a **systematic or methodical way** and **individually accessible** by electronic or other means.

Four cumulative criteria:

1. **Collection**: A collection or compilation of data.
   - Recital 17 DD: Data could be in the form of text, sound, images, numbers, facts etc.



Subject-matter

2. **Independence of data:** Data must constitute separate and independent items.

- Autonomous informative value of data (as individual items or a few items of data in combination with each other) .
- *Test:* Item(s) of data must have independent value when separated from database (CJEU, C-444/02 Fixtures Marketing [2004] para. 33)

3. **Systematic/methodical arrangement:** Data must be arranged in a systematic or methodical way (arrangement is logical not random).

4. **Individual accessibility**: Independent units of data can be retrieved individually.

- Collection should include technical or other means (e.g. electro-optical processes, an index or a particular plan or method of classification) to allow retrieval of independent items of data. (CJEU, C-444/02 Fixtures Marketing [2004] para. 30)



Key Features of a database

- ✓ Collection or compilation of data.
- ✓ Data constitute separate and independent items (each item of data can have independent informative value on its own).
- ✓ Data is arranged in a systematic or methodical way (arrangement is logical not random).
- ✓ Independent items of data can be retrieved individually (independently accessible).



Quiz 1: Database or not?

E-version of *De Telegraaf* newspaper



The movie “The Matrix” (as when you watch it in the cinema)



A map of Den Bosch as displayed on Google maps.

Your Facebook page

Data stored on the Ryanair website



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Why cannot movies, songs, novels etc. qualify as databases?

Recital 17 DD:

Whereas the term 'database should be understood to include literary, artistic, musical or other collections of works or collections of other material such as texts, sound, images, numbers, facts, and data; whereas it should cover collections of independent works, data or other materials which are systematically or methodically arranged and can be individually accessed; whereas this means that a recording or an audiovisual, cinematographic, literary or musical work as such does not fall within the scope of this Directive.

Which of the following can be protected under IP law? Why?

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Table: customers

customer_id	first_name	last_name	phone	country
1	John	Doe	817-646-8833	USA
2	Robert	Luna	412-862-0502	USA
3	David	Robinson	208-340-7906	UK
4	John	Reinhardt	307-242-6285	UK
5	Betty	Taylor	806-749-2958	UAE



3.6 kg



Only products of (human?) intellect can be protected by IP law

Objective & Structure

Objective: Examine and analyze database protection under copyright and the sui generis database right as provided by the EU Database Directive [DD] (1996).

Structure:

1. We will focus on the fundamental aspects of copyright protection and sui generis database rights protection for databases.
   i. Criteria for protection.
   ii. Ownership and Duration of protection.
   iii. Scope of protection.

2. Quick Look: Exclusion of machine-generated databases (MGDs) from the scope of the sui generis database right.

Copyright protection of Databases

Criteria

Art. 3(1) DD, Sole criteria: **Originality**

The database must be original → The selection **AND/OR** arrangement of database must qualify as being its the **A**uthor's **O**wn **I**ntellectual **C**reation (***AOIC*** standard of originality).

How to test whether the AOIC standard of originality is fulfilled.

- The contents of the database have been selected and/or arranged according to the database-maker's personal creative choices leading to his personal touch being stamped on the database.

  - Creator has had the freedom to engage in his own (arbitrary) creative decision-making in selecting and/or arranging the contents.
    - AOIC standard will **not** be fulfilled when the selection and arrangement of the database is dictated by technical of functional considerations, rules or restrictions thereby leaving no space for the exercise of free and creative choices. E.g. industrial standards, organizational constraints.

  - Creator has used this freedom to **make personal creative choices** in selecting and/or arranging the contents resulting in the database being stamped with the creator's personal touch.
    - Mere expenditure of **skill, labour and know-how** is **insufficient** for fulling AOIC standard of originality!

Criteria: Checklist

Does the database fulfill the AOIC standard of originality in the selection OR arrangement of contents?

Selection of contents in database

- ✓ Creator has had freedom to make his personal creative choices in selecting contents.

  Yes ↓ No 

- ✓ Creator has used this freedom to make his personal creative choices in selecting contents.

  Yes ↓ No 

- ✓ By reason of the creator making his personal creative choices the selection of the contents of the database is stamped with the creator's personal touch.

  Yes  No 

Arrangement of contents in database

- ✓ Creator has had freedom to make his personal creative choices in arranging contents.

  Yes ↓ No 

- ✓ Creator has used this freedom to make his personal creative choices in arranging contents.

  Yes ↓ No 

- ✓ By reason of the creator making his personal creative choices the arrangement of the contents of the database is stamped with the creator's personal touch.

  Yes  No 

Ownership

Art. 4

- Author: Natural person who carried out selection and arrangement of contents, OR
- Employer of that natural person, which could be a natural or legal person (e.g., company) when the law provides that the copyright is assigned to the employer (e.g., contract of employment).



Nationality, residence (whether in the EU or outside) is not bar to obtaining copyright protection or ownership of copyright. E.g., A Chinese citizen who is employed by an EU company (but works remotely and never comes to the EU) creates a database while living in Beijing. This database can still be protected by copyright in the EU if the relevant criteria are fulfilled.

Duration

Database protected during the lifetime of author (natural person who carried out selection and arrangement of contents) + 70 years.

➢ If co-authored, 70 years after the death of the last surviving author.



Can the following obtain copyright protection?

A. Louvre Museum collection

The **Louvre Museum** in France has over 300 renaissance portrait paintings that were created during the 15th-16th Centuries. The Museum has digitized all these paintings. The resulting digital images are stored in the Museum's digital archives.

Vincent is a French citizen who is employed as a curator at the Louvre Museum. **Rajiv** is an Indian citizen who is studying art in Paris. During the Summer holidays he is working as an intern at the Louvre Museum under Vincent's supervision.

Vincent wishes to create a compilation of digital images of the best renaissance portrait paintings held by the Museum. He asks Rajiv to assist him in the task.

- Rajiv evaluates all digital images of renaissance portrait paintings stored in the Museum's digital archives and **selects** digital images of 100 paintings which in his opinion have the highest artistic merit and beauty.
- Rajiv **categorizes** the 100 digital images into five themes based on the subjects depicted in the portraits (i.e., "Lovely Ladies", "Princes & Warriors", "Merchants & Adventurers", "Saints & Men of God", "Cherubic Children"). The lovely Ladies category is the largest (30 images) while the Cherubic Children category is the smallest (15 images).
- Vincent **sequences** the digital images within each theme according to his own personal creative preferences.

The thematically arranged digital images are published on the Louvre Museum Website as a digital database entitled the "100 best renaissance portrait paintings".

Answer

Step 1: Is it a database within the definition in Article 1(2) of DD?

➢ Definition : A collection of independent works, data or other materials arranged in a systematic or methodical way and individually accessible by electronic or other means.

Step 2: If answer to Step 1 is yes, can database obtain protection under copyright?

➢ Test in Article 3(1) of DD: Does selection/arrangement of database fulfill AOIC standard of originality by qualifying as its author's own intellectual creation?

Q.1: If it can be protected by copyright, who is the author of the database?

Q.2: Who do you think will own copyright in the database?

B. Viewership data

- **Simson** is a Dutch company registered in the Netherlands. Its registered office and principal place of business is in Eindhoven.
- Simson spends 20 million EUR per year to purchase viewership data (i.e., data on audience sizes) for TV-programs broadcast in different EU Member States from various media research companies. Simson spends an extra 5 million EUR per year to check the accuracy of the data it purchases.
- Simson's employees expend much skill and labor in systematically arranging this data according to the language of the TV program to form a database (e.g., the data is categorized as English language TV-programs, Spanish language TV-programs, Dutch language TV-programs etc.).

Answer

Step 1: Is it a database within the definition in Article 1(2) of DD?

➢ Definition : A collection of independent works, data or other materials arranged in a systematic or methodical way and individually accessible by electronic or other means.

Step 2: If answer to Step 1 is yes, can database obtain protection under copyright?

➢ Test in Article 3(1) of DD: Does selection/arrangement of database fulfill AOIC standard of originality by qualifying as its author's own intellectual creation?

Scope

Art. 5 DD: Engaging in following acts (for commercial/ non-commercial purposes) require copyright owner's authorization:

a) Temporary/ permanent reproduction of database in whole or in part, by any means and in any form (directly/indirectly). [Does not cover mere consultation of a database made available to the public].

b) Distribution (sale) of database or its copies to the public.

c) Adaptation or modification of database (e.g. translation).

d) Communication, display or performance of the database to the public. [Communication: transmission of the database by wire/wireless means including the making available to the public in such a way that members of the public may access them from a time and place individually chosen by them i.e., interactive on-demand transmissions].

- **Sharing hyperlinks and requirements of a "new public"**: CJEU, C-466/12, *Svensson (*2014): Provision of clickable links to a protected work would not be considered an infringement of a rightholder's exclusive right to communicate the work to the public, unless access to the protected work had been restricted to certain groups of authorised users. Situation would be different where a hyperlink makes it possible for users of the site on which that link appears to circumvent restrictions put in place by the site on which a protected work originally appeared, such as the restriction of public access to the website's paying subscribers only.

Infringement test: Copyright.

- ✓ Database is protected by copyright (AOIC standard of originality is fulfilled; protection has not expired).

  Yes → 

  No → 

- ✓ The intended acts come within the scope of copyright in the database.

  Yes → 

  No → 

- ✓ You have authorization of the copyright owner to carry out the intended acts (e.g., license) OR the intended acts fall within a legally granted exception to copyright in the database.

  Yes → 

  No → 



Do the unauthorized act(s) come within the scope of copyright in the database? Is there *potential* infringement?

Louvre Museum collection

The “100 most beautiful renaissance portrait paintings” digital database is made available free of charge on the Louvre Museum’s website, **but only to those who have registered themselves as users of the Louvre Museum website.**

Jan is an Art Historian living in Amsterdam.

- Jan copies the 15 images of the “Cherubic Children” category of the “100 most beautiful renaissance portrait paintings” onto his own computer and arranges them in the same way they were arranged by Vincent within this category.

  Direct reproduction of a part: Reproduction right

- He prints them as a booklet entitled “Renaissance Child Portraits” which he sells to the public at 10 euros a booklet. The booklet includes information on each painting and his own critical commentary on why each painting constitutes a masterpiece of child portraits.

  Adaptation right + Distribution right

An electronic low-resolution version of the booklet is made available on his personal website free-of-charge, so that any member of the public can access his booklet.

Communication to the public right

Sui generis protection

Preliminary condition: Who can benefit from protection?

Article 11 DD: ***Sui generis right can only be obtained if the maker of the database is:***

- Natural person: Who holds nationality of an EU Member State or maintains habitual residence in the territory of the EU.

- Legal person: A company or firm that fulfills ***both*** of following criteria.

  1. Formed in accordance with the law of an EU Member State and having its registered office within the community.

  AND;

  2. Having its central administration **or** principal place of business within the Community. [Because operations of company must be genuinely linked on an ongoing basis with the economy of an EU Member State].



Can the following persons/entities potentially benefit from the *sui generis* database right?

Can the following persons/entities potentially benefit from the *sui generis* database right?

- Vanson is an Irish company registered in Dublin. Its registered office and principal place of business is in Amsterdam. It has corporate offices in Berlin, Mumbai and Mexico City.
- Yin-Shi is registered as a company in China. It maintains its registered office in Beijing but has corporate offices in Paris, London and New York.
- Vincent is a French citizen who maintains his habitual place of residence in India.
- Sheila is a Nigerian national. She is registered as a BSc student at the University of Tilburg but follows her lectures over Zoom from Lagos, Nigeria. Each year, she will come to Tilburg for one month to complete her exams.

Criteria

Art. 7 DD: There must have been a **substantial** investment made in either the obtaining, verification or presentation of contents of database.

- Obtaining: investment made in seeking out and collecting existing data (even if the initial objective of collecting the data was not for creating the database per se).
- Verification: investment made in ensuring the accuracy/reliability of the data during collection and afterwards (periodic monitoring the 'materials' of a database to ensure completeness and accuracy).
- Presentation: Selection and/or arrangement of contents in a way that is individually accessible.

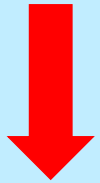
Investment: Can be financial, human effort, expertise, time, know-how, skill and labor etc.

Substantial: Depends on whether the investment is substantial for such a database in the specific field of industry (decided on a case-by-case basis).

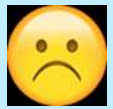
Criteria: Checklist

I. Preliminary condition: Can creator benefit from protection under Article 11 DD?

Yes



No



II. Has there been a substantial investment in **one or more** of the following as per Article 7 DD?

Obtaining contents?

Yes 

No 

Verification of contents?

Yes 

No 

Presentation of contents?

Yes 

No 

Duration

Art. 10 DD:
Database protected for 15 years, calculated from 1 January of the year *after* date of completion of database.

- **Art. 10(3) DD**: 'rolling' *sui generis* right: Any 'substantial change' (i.e., additions, deletions, re-arrangement, verifications to the contents of a database) that requires a substantial new investment evaluated qualitatively or quantitatively shall qualify the database as a whole to enjoy a new period of protection.

Ownership

Art. 7 DD: Natural person or legal person who is responsible for the substantial investment made in the creation of the database.



Can the following obtain protection under the *sui generis* database right?

Viewership data

- **Simson** is a Dutch company registered in the Netherlands. Its registered office and principal place of business is in Eindhoven.
- Simson spends 20 million EUR per year to purchase viewership data (i.e., data on audience sizes) for TV-programs broadcast in different EU Member States from various media research companies. Simson spends an extra 5 million EUR per year to check the accuracy of the data it purchases.
- Simson's employees expend much skill and labor in systematically arranging this data according to the language of the TV program to form a database (e.g., the data is categorized as English language TV-programs, Spanish language TV-programs, Dutch language TV-programs etc.).

Answer

Step 1:Is it a database within the definition in Article 1(2) of DD?

➢ Definition : A collection of independent works, data or other materials arranged in a systematic or methodical way and individually accessible by electronic or other means.

Step 2:Does Simson's database fulfill the preliminary criteria (under Article 11 of DD) to qualify for *sui generis* database protection?

- ➢ Natural person: EU citizen or habitual residence in EU.
- ➢ Legal person: Formed in accordance with law of EU MS + registered office in EU + central administration **or** principal place of business in EU

Step 3: If answer to Step 2 is yes, can the database obtain *sui generis database* protection?

- ➢ Art. 7 DD: There must have been a **substantial investment** made in either the **obtaining, verification or presentation** of contents of database.

Difference between creation and obtaining of data

Investment made in creating (generating) data or verifying data at the point of creation is **not relevant** in determining whether a substantial investment has been made in obtaining or verifying contents of database!

Objective: “Spin-off” databases (i.e., compilations of data that are generated as by-products ('spin-offs') of other activities cannot claim sui generis database protection on the basis of having made an investment in the obtaining (i.e., collection) of data, because the investment is not made in obtaining existing data but in engaging in a different activity as a result of which the data is generated.

E.g., A TV program schedule that is generated by a TV station, a database of flight schedules generated by an airline company.

But,

Observed data (e.g. compilation of data about football matches in progress including goals and goalscorers, yellow and red cards and which players were given them) is obtained data, not created data. [Football Dataco & Others v Stan James [2013] England & Wales Court of Appeal Civ 27].



Is the investment in the creation or obtaining of data?

British Horseracing Board (BHB) database [Case C-203/02, *BHB v William Hill* (2004)]

BHB organizes horse-races in the UK. In order to, organize these races it registers information concerning owners, trainers, jockeys and horses, decides on the horses to be entered for each race (based on their weight, handicapping etc.) and finally it compiles the lists of horses running in the races. This requires a large financial investment (approx. £4 million per annum).

The data is entered into the BHB database.

N.B. This case took place before Brexit, so BHB fulfills the preliminary condition under Article 11 DD.

Scope

Art. 7 DD: Engaging in following acts (for commercial/ non-commercial purposes) require sui generis right owner's authorization:

- Extraction and/or re-utilization of **whole** or **substantial part** of contents of database evaluated qualitatively and/ or quantitatively.

Extraction ⟷ Reproduction (copying)

Appropriation and transfer of contents of database from one medium to another (i.e. direct/indirect reproduction).

➢ Does not require physical copying. Transferring the elements of one database to another database following on-screen visual consultation of the first database will constitute extraction. (CJEU, C-304/07, *Directmedia Publishing GmbH v Albert-Ludwigs-Universität Freiburg* [2008]).

- **Re-utilization** ⟷ Any form of making available contents (either in their original form or in modified form) of a database to the public by communication, display or performance or by distribution (sale).

Scope

How to test "Substantiality

- "Quantitative: Volume of data assessed in relation to total volume of contents of database.
- Qualitative: Scale of investment in obtaining, verification and presentation of contents extracted/re-utilized is substantial.

Includes, **repeated and systematic** extraction and/or re-utilization of insubstantial parts of the contents of database where this is **contrary to the normal exploitation of that database or unreasonably prejudices the legitimate interests of the producer of the database.** [Art. 7(5) DD].

Infringement test: Sui generis database right.

- ✓ Database is protected by the sui generis database right (preliminary condition + criteria for protection fulfilled; protection has not expired).

  Yes → 

  No → 

- ✓ The intended acts come within the scope of the sui generis database right.

  Yes → 

  No → 

- ✓ You have authorization of the sui generis database rights holder to carry out the intended acts (e.g., license) OR the intended acts fall within a legally granted exception to the sui generis database right.

  Yes → 

  No → 

- ✓ You extract/re-utilize a substantial part of the contents of the database OR You extract/re-utilize repeatedly and systematically insubstantial parts of the contents of database in a way that is contrary to the normal exploitation of that database or unreasonably prejudices the legitimate interests of the producer of the database.

  Yes 

  No 



Does the unauthorized act come within the scope of the sui generis database right? Is there *potential* infringement?

“Romanian-language TV programs” is the smallest category of data in Simson’s database. It comprises less than 10% of the total volume of the database. However, obtaining and verifying the data in this category was very expensive and accounts for more than 35% of the total investment made in the creation of the database.

Jan has access to Simson’s database. Without Simson’s authorization, Jan copies all the data included in the category of “Romanian-language TV programs” onto his own hard-drive.

Does Jan’s act infringe on the sui generis database right?

a. Yes. Because he has re-utilized a substantial part of the database.

b. No. Because he has only re-utilized an insubstantial part of the database.

c. Yes. Because he has extracted a substantial part of the contents of the database.

d. No. Because he has only extracted an insubstantial part of the database.



Exam Practice during Lecture 6

DataMed's database

DataMed is a Dutch company having its registered office and principal place of business in Amsterdam. In 2021 it spent approx. 20 million euros to collect patient data from hospitals across the world on several debilitating diseases (e.g. Alzheimer's disease, Parkinson's disease and Multiple Sclerosis). Approx. 8 million was spent in obtaining data on Alzheimer's disease.

DataMed's employees, arrange this data in an electronic database according to an internationally accepted organizational system that enables the data to be easily analyzed and retrieved by medical researchers. This is a task that requires much expenditure of skill, labour and expertise.

Questions

1. Can DataMed's database qualify as a database under the definition provided in Article 1(2) of EU Database Directive (1996)? Yes? No? Explain the reasons for your answer [7 points].

2(a). *Regardless of your answer to Q.1 assuming that DataMed's database does qualify as a database Article 1(2) of Database Directive (1996):* Can DataMed's database obtain copyright protection? Yes? No? Explain the reasons for your answer [6 points].

2(b). *Regardless of your answer to Q.1 assuming that DataMed's database does qualify as a database Article 1(2) of Database Directive (1996):* Can DataMed's database obtain protection under the *sui generis* database right? Yes? No? Explain the reasons for your answer [9 points].

Question 1: **Can DataMed's database qualify as a database under the definition provided in Article 1(2) of EU Database Directive (1996)?**
Yes? No? Explain the reasons for your answer [7 points].

- **Yes [1 point].**
- Patient data constitute **separate and independent [1 point]** items, because even data belonging to one single patient can have **autonomous informative alue [1 point]** when separated from the database.
- Patient data is arranged in a **systematic and methodical** way **[1 point] according to an internationally accepted organizational system that enables the data to be easily analyzed [1 point].**
- Patient data is **independently accessible [1 point]** because they can be **retrieved individually [1 point].**
- Therefore, DataMed's database qualifies as a database under the definition provided in Article 1(2) of the EU Database Directive (1996).

Question 2(a): **Can DataMed's database obtain copyright protection?**
Yes? No? Explain the reasons for your answer [6 points].

- **No [1 point].**
- Under Article 3 of the Database Directive, the sole criterion for obtaining copyright protection is that the selection and/or arrangement of the database fulfills the **AOIC standard [1 point]** of **originality [1 point]**. The AOIC standard of originality is fulfilled when the creator(s) of the database has engaged in their personal creative choices in the selection and/or arrangement of the data in the database resulting in the database being stamped with their personal touch stamp.
- There is no evidence that there has been a selection of data. So, the question is whether the arrangement of the database can fulfill the AOIC standard of originality.
- As per the given facts, the arrangement of the data will not fulfill the AOIC standard of originality because the **arrangement is done according to an internationally accepted organizational system [1 point]**, which has the **functional purpose [1 point]** of enabling the data to be easily analyzed and individually retrieved.
- This means that DataMed's employees did not have the freedom to exercise their **personal creative choices** in the arrangement of the data in a way that would allow the database to reflect their **personal touch stamp**. **[1 point if at least one set of keywords are present]**
- Therefore, the AOIC standard of originality is not fulfilled, and DataMed's database will not obtain copyright protection.

Question 2(b): **Can DataMed's database obtain protection under the sui generis database right?**
Yes? No? Explain the reasons for your answer [9 points].

- **Yes [1 point].**

- The preliminary requirements under Article 11 of the Database Directive are fulfilled because, DataMed is a company that has been **formed in accordance with Dutch law** and **has its registered office [1 point if either is present]** and **principal place of business [1 point] in Amsterdam/Netherlands** which is in the **EU [1 point if either is present].** So, the database can obtain protection under the *sui generis* database right if the criteria for protection under Article 7 of the Database Directive are fulfilled.

- DataMed can be considered to have made a **substantial investment [1 point]:**

  - in **obtaining [1 point]** the data because **20 million euros is a sizeable sum of money** and **the database is stated as being one of the most expensive databases on debilitating diseases in the EU [2 points if either is present]**

  OR

  - in the **presentation [1 point]** of the data **because its employees expend much skill, labour and expertise [2 points]** in arranging the data according to an internationally accepted organizational system.

- Therefore, the DataMed's database can obtain protection under the s*ui generis* database right provided in Article 7 of the EU Database Directive (1996).

Machine-generated databases

Machine-generated dynamic databases

Databases that are created, updated, and managed primarily by automated processes and machines (including the IoT).

Some Characteristics:

- Dynamic: Data is continuously or periodically collected, modified and/or updated (high velocity, real-time). The structure of the database can also be subject to change.
- Arrangement: Can be relational, non-relational or hybrid.

Q: Do you think a non-relational database could qualify as a database as per the definition provided in Article 1(2) DD?

Can machine-generated dynamic databases obtain copyright protection?

Art. 3(1) DD: Sole criterion is AOIC standard of **Originality**

- Creator has made their **personal creative choices** in selecting and/or arranging the contents resulting in the database being **stamped with the creator's personal touch.**

Machine-generated dynamic database:

- Selection and arrangement of data automated and performed by algorithms.
  - Human contribution is usually merely technical or facilitative being limited to establishing the technological infrastructure by creating and programming the algorithm (Lukas Johnson-Hecker, 2020, UvT).

- Selection and arrangement of data determined by technical/functional considerations.

Sui generis database right?

Difficult to distinguish between ‘creating’ and ‘obtaining’ when big data is generated, aggregated, processed and exchanged in real-time.

- Gervais: Big Data corpora are neither “obtained” nor “collected”; they are generated by the machine. [Gervais, 2019, 29].
  - Can they be spin-off databases i.e., data is generated in the course of some other function (e.g., agricultural machinery that records data while performing another function)?

- But, sometimes machine-generated databases generate data based on observed occurrences.
  - Data from an automatically administered database can be deemed as ‘obtained’ when it records a physical or digital, objectively appreciable occurrence (i.e., data observed and recorded by a satellite).

Substantial investment in presentation and verification?

Leistner: Investment into the establishment of an infrastructure for obtaining, verifying or presenting pre-existing data could be relevant for assessing substantiality of the investment [Leistner, Big Data and the EU Database Directive 96/9/EC: Current Law and Potential for Reform by Matthias Leistner :: SSRN, p.2]

- Could substantial investments in developing a technological (e.g., AI-based) infrastructure for aggregating, cleaning and organizing the data in a machine-generated dynamic database be considered a substantial investment in the verification and presentation of data?

Substantial investment in new versions?

Dynamic databases are subject to a constant process of updating in order to integrate new data inputs. Results in multiple new versions of database.

Art. 10(3) DD: 'rolling' *sui generis* right: Any 'substantial change' (i.e., additions, deletions, re-arrangement, verifications to the contents of a database) that requires a substantial new investment evaluated qualitatively or quantitatively shall qualify the database as a whole to enjoy a new period of protection.

- Can there be a 'substantial change' requiring a substantial investment each time a machine-generated database is updated?

EU Data Act

Objective: To boost the EU's data economy by removing barriers to accessing personal data [subject to GPDR rules and other laws] and non-personal data and by fostering a competitive and reliable European cloud market.

Article 3: Obligation to make data accessible to user

- Connected devices and related services (i.e., machine-generated data) should be designed and provided in a manner that the 'product data' and 'related service data' (including related metadata necessary to interpret and use those data) are by default easily, securely, free of charge, in a comprehensive, structured, commonly used and machine-readable format, and, where relevant and technically feasible, directly accessible to the user.

Article 4: Right of user to access data

- Where data cannot be directly accessed by the user from the connected product or related service, data holders shall make readily available data, as well as the relevant metadata necessary to interpret and use those data, accessible to the user easily, securely, free of charge, in a comprehensive, structured, commonly used and machine-readable format and, where relevant and technically feasible, continuously and in real-time. This shall be done on the basis of a simple request through electronic means where technically feasible.

Article 5: Right of user to share data with third parties

- Upon request by a user, or by a party acting on behalf of a user, the data holder shall make available readily available data, as well as the relevant metadata necessary to interpret and use those data, to a third party without undue delay, of the same quality as is available to the data holder, easily, securely, free of charge to the user, in a comprehensive, structured, commonly used and machine-readable format and, where relevant and technically feasible, continuously and in real-time.

Machine-generated databases excluded from protection under the sui generis database right

CHAPTER X

SUI GENERIS RIGHT UNDER DIRECTIVE 96/9/EC

Article 35

Databases containing certain data

The *sui generis* right provided for in Article 7 of Directive 96/9/EC shall not apply when data is obtained from or generated by a product or related service falling within the scope of this Regulation, in particular in relation to Articles 4 and 5.

Recital (84) In order to eliminate the risk that holders of data in databases obtained or generated by means of physical components, such as sensors, of a connected product and a related service or other machine-generated data, claim the sui generis right under Article 7 of Directive 96/9/EC, and in so doing hinder in particular the effective exercise of the right of users to access and use data and the right to share data with third parties under this Regulation, it should be clarified that the sui generis right does not apply to such databases. That does not affect the possible application of the *sui generis* right under Article 7 of Directive 96/9/EC to databases containing data falling outside the scope of this Regulation provided the requirements for protection in accordance with Article 7(1) of that Directive are fulfilled.

Additional:
Excerpts from Data Act

Promoting access and use of machine-generated data

CHAPTER II

BUSINESS TO CONSUMER AND BUSINESS TO BUSINESS DATA SHARING

Article 3

Obligation to make product and related service data accessible to the user

1. Connected products shall be designed and manufactured, and related services shall be designed and provided, in such a manner that product data and related service data, including the relevant metadata necessary to interpret and use the data, are, by default, easily, securely, free of charge, in a comprehensive, structured, commonly used and machine-readable format, and, where relevant and technically feasible, directly accessible to the user.

Article 2

Definitions

(1f) 'product data' means data, generated by the use of a connected product, that the manufacturer designed to be retrievable, via an electronic communications service, a physical connection or on-device access, by a user, data holder or a third party, including, where relevant, the manufacturer;

(1fa) 'related service data' means data representing the digitisation of user actions or events related to the connected product, recorded intentionally by the user or as a by-product of the user's action, which is generated during the provision of a related service by the provider;

(2) 'connected 'product' means an item, that obtains, generates or collects, data concerning its use or environment, and that is able to communicate product data via an electronic communications service, a physical, connection or on-device access and whose primary function is not the storing, processing or transmission of data on behalf of third parties, other than the user;

(3) 'related service' means a digital service other than an electronic communications service, including software, which is connected with the product at the time of the purchase in such a way that its absence would prevent the product from performing one or more of its functions, or which is subsequently connected to the product by the manufacturer or a third party to add to, update or adapt the functions of the product;

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Users have right to access machine-generated data and to share with third parties (empowering users)

Article 4

The rights and obligations of users and data holders to access, use and make available product and related service data

1. Where data cannot be directly accessed by the user from the connected product or related service, data holders shall make readily available data, as well as the metadata that is necessary to interpret and use that data, accessible to the user without undue delay, easily, securely and in a comprehensive, structured, commonly used and machine-readable format, free of charge and, where relevant and technically feasible, of the same quality as is available to the data holder, continuously and in real-time. This shall be done on the basis of a simple request through electronic means where technically feasible.

Article 5

Right of the user to share data with third parties

1. Upon request by a user, or by a party acting on behalf of a user, the data holder shall make available readily available data, as well as the metadata that is necessary to interpret and use that data, to a third party, without undue delay, free of charge to the user, of the same quality as is available to the data holder, easily, securely, in a comprehensive, structured, commonly used and machine-readable format and, where relevant and technically feasible, continuously and in real-time. The making available of the data by the data holder to the third party shall be done in compliance with Articles 8 and 9.

Article 2

Definitions

(1e) 'readily available data' means product data and related service data that a data holder lawfully obtains or can lawfully obtain from the connected product or related service, without disproportionate effort, going beyond a simple operation;